

UNSTOPPABLE — DAY 10

Advanced Lookup Functions in Excel

Learning status: Concepts introduced. Hands-on practice and troubleshooting will be revisited because some formulas did not work in the current Excel setup.

1. INDEX()

Purpose: Returns a value from a range based on its relative row and column position.

Syntax: =INDEX(array, row_num, [column_num])

Example table (A2:C4):

Employee	Department	Salary
Rahul	Sales	35000
Priya	Finance	42000
Amit	Marketing	38000

=INDEX(A2:C4,2,3) → 42000 (Priya's salary).

=INDEX(A2:C4,3,2) → Marketing.

Row and column positions are counted within the selected range, not by worksheet row numbers.

2. MATCH()

Purpose: Finds a value and returns its relative position in a range.

Syntax: =MATCH(lookup_value, lookup_array, match_type)

Use 0 for an exact match.

=MATCH("Priya",A2:A4,0) → 2

=MATCH("Amit",A2:A4,0) → 3

MATCH returns a position within the selected range, not the worksheet row number or salary.

3. INDEX + MATCH

MATCH finds a position; INDEX returns the value at that position.

Find Priya's salary: =INDEX(C2:C4,MATCH("Priya",A2:A4,0)) → 42000

Find Amit's department: =INDEX(B2:B4,MATCH("Amit",A2:A4,0)) → Marketing

The lookup column and return column can be separate ranges; the lookup column does not need to be leftmost.

4. XMATCH()

Purpose: Finds a value and returns its relative position; a more flexible alternative to MATCH.

Syntax: =XMATCH(lookup_value, lookup_array, [match_mode], [search_mode])

=XMATCH("Priya",A2:A4) → 2

XMATCH uses exact matching by default and supports reverse search. It is not available in every Excel version; older versions may return #NAME?.

5. Multiple-condition lookups

Use multiple conditions when one field alone is not unique, such as employee name plus department.

Employee	Department	Salary
Rahul	Sales	35000
Priya	Finance	42000
Rahul	Marketing	48000
Amit	Sales	38000

Modern Excel array formula: =INDEX(C2:C5,MATCH(1,(A2:A5="Rahul")*(B2:B5="Marketing"),0))

Expected result: 48000. Multiplying the two TRUE/FALSE tests means both conditions must be TRUE. Compatibility may vary by Excel version.

Helper-column alternative

In D2: =A2&"|"&B2, then fill down to create keys such as Rahul|Marketing.

Lookup formula: =INDEX(C2:C5,MATCH("Rahul|Marketing",D2:D5,0)) → 48000

This approach is often easier to troubleshoot in older Excel versions.

6. Lookup error handling

If MATCH or XMATCH cannot find a value, it usually returns #N/A. Check spelling, spaces, data types, and the lookup range.

Example: =IFERROR(INDEX(C2:C4,MATCH("Nina",A2:A4,0)),"Not found")

If Nina is not in A2:A4, the formula returns "Not found". IFERROR can also hide unrelated errors, so use it thoughtfully.

7. Troubleshooting checklist

- Check whether the function is supported by your Excel version (XMATCH is unavailable in some older versions).
- If Excel shows #NAME?, check function availability and spelling.
- If Excel shows #N/A, verify the lookup value exists and check spaces, spelling, and data types.
- If the formula displays as text, change the cell format to General and re-enter it; check for a leading apostrophe.
- If your regional settings require semicolons, replace argument-separating commas with semicolons.
- For multiple-condition lookups, try the helper-column approach if the array formula fails.
- Test each component separately: MATCH first, then INDEX, then combine them.

8. Key takeaways

- INDEX retrieves a value by position.
- MATCH finds a relative position; use 0 for exact matching.
- INDEX + MATCH performs a flexible lookup across separate ranges.

- XMATCH is more flexible but requires a compatible Excel version.
- Multiple-condition lookups require all criteria to match the same record.
- IFERROR can make missing lookups clearer, but should not conceal unrelated problems.